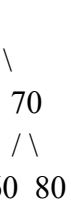
Set A

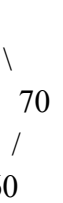
Instructions

1. **Write pseudocode only — do not write in any programming language.**
2. **If you write pseudocode, it must be neatly written and include comments explaining steps.**
3. **If you choose to write steps/approach, write them point by point, clearly.**
4. **First write the general pseudocode/approach, then apply it on the given example to check whether it works, and finally write the output.**
5. **If you do not follow these instructions, your answer will not be evaluated.**

Q1. A company stores employee IDs in a **binary tree**. To prepare a sorted report, the manager asks you to print the IDs in **increasing order**. Which tree traversal should you use? Write the pseudocode/steps.



Q2. A library stores book IDs in a **BST**. Current tree (book IDs inserted in this order): **50, 30, 70, 20, 40, 60**



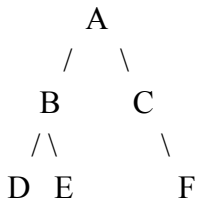
Now, a new book with ID **65** arrives. Write the steps/pseudocode to insert it in the BST, and draw the **final tree**.

Set B

Instructions

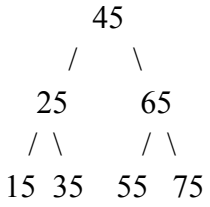
1. **Write pseudocode only — do not write in any programming language.**
2. **If you write pseudocode, it must be neatly written and include comments explaining steps.**
3. **If you choose to write steps/approach, write them point by point, clearly.**
4. **First write the general pseudocode/approach, then apply it on the given example to check whether it works, and finally write the output.**
5. **If you do not follow these instructions, your answer will not be evaluated.**

Q1. A teacher wants to list a project hierarchy.
She needs to print the **head first, then its left sub-part, then right sub-part, then their sub-parts**.
This is the given tree of project parts:



1. Which traversal will you use?
2. Write the pseudocode/steps.
3. Write the **output sequence**.

Q2. A student database stores IDs in a **BST**.
Rule: **Insert only if the ID is odd. If it is even, ignore it.**
Current tree (IDs inserted in this order): **45, 25, 65, 15, 35, 55, 75**



Now, four new student IDs arrive: **37, 88, 7** and **40**.

1. Write the pseudocode/steps to insert according to the rule.
2. Draw the **final tree**.